Topic : Anticoccidial drug resistance in fowl coccidia

1. Rational

Coccidiosis is among the most common diseases of poultry caused by protozoan parasite of Emeria species in spite of advances made in prevention and control through chemotherapy, management and nutrition. It causes massive destruction of the epithelial cells and characterized by droopiness, paleness of the comb, diarrhea and occasional appearance of blood in droppings. The death rate can be quite high, both in chicks and in adults (Fanatico, 2006).

Antimicrobial resistance (AMR) has its origin in the unregulated overuse and misuse of antimicrobials in the humans, terrestrial and aquatic animals and in production sectors, exerting selection pressure on pathogen populations that encourages the development of resistance and exchange of resistance genes. A large number of antimicrobials are used in the animal industry in Asia, and AMR is widely prevalent.

Weak or non-existent regulatory frameworks governing antimicrobial use (AMU), sub-optimal enforcement and compliance with existing guidelines, low levels of AMR awareness, and inadequate commitment to responsible antimicrobial stewardship are driving development of AMR. Regionally, AMR mitigation measures in the food and agriculture realm lag behind the human health sector. There are gaps in data and information across the food, agriculture and other sectors on AMU at farm settings and beyond.

These hamper the assessment of AMU and therefore create difficulties in determining levels and flows of residue and AMR across the food value chain system to the point of consumption.

Therefore, the research is able to support identification of the best and the worst practices in using antimicrobials, the adapted set of interventions that could changes stakeholder's behaviors, providing good knowledge and practices for animal farm levels, environmental persons and in drug sellers, and then design and implement a training module on AMU for all stake holders.

Control of coccidiosis chiefly depends upon prophylactic chemotherapy with anticoccidial drugs (McDougald and Reid, 1994). With the passage of time, a large number of anticoccidial drugs have been introduced. Periodic monitoring of coccidia for development of resistance against commonly used prophylactics is, therefore, imperative for economical poultry production. Better awareness and socio-economic conditions of farmers and state-sponsored animal health facilities in industrialized or developed countries have led to frequent reports on anticoccidial resistance (Stephan et al.,1997). In the case of poultry reared for meat production, anticoccidial drugs are frequently included in the feed for the main tenancies of the bird, and this ensures that few parasites escape the effects of medication. This practice has persistently resulted in the development of resistance (Chapman, 1997). This paper discusses the different aspects of the drug resistance in the following sections for the better understanding and control of resistant Eimeria species.

2. Literature review

2.1 Livestock production in Laos.

The strategy for an agricultural development plan for 2020 to 2025 indicated that the local breeds of livestock production system should shift from the traditional to a modernized production system by crossing with foreign purebreds to increase productivity, disease resistance and be market oriented. Lao government continuously supports, protects, and promotes the livestock and poultry industries. Almost 98% of livestock farmers in Laos own at least one local breed of cattle, buffalos, pigs, and goats. Throughout 45 years, Livestock Research Center (LRC) has been thriving to improve and develop local breeds of livestock and poultry (cattle, pigs, goats, and chickens) by enhancing their productive and reproductive performance (MAF, 2015).

The overall livestock population in Laos has been increased year-by-year for 45 years until now. However, it experienced a decrease of nearly 22% in 2000 compared to 1995. This is because of the new national strategy on economic revolution, especially a free zone for crop production, but rose again until 2019. There was only a 6.58% increase in 1985 due to the shift-point of new national economic revolutions. A more than 5 times increased (32%) was observed five years later (Table 1). Then comes a slight decrease in 2015 because of more rubber, cassava, and other cash crop production (MAF, 2020).

2.2. Antimicrobial used and antimicrobial resistant

The rapid intensification of the livestock sector in Southeast Asia has been found to be associated with an expanding use of medicines, because of those animal farms are facing several challenges in any aspects, especially infectious disease outbreak including microbes and parasites. Therefore, antimicrobials are used widely for the treatment, prevention of infectious disease, and growth promotion effective to rise of AMR in both humans and livestock (Putthana, 2022).

An estimated 131.109 tons of all antimicrobials were used animal feeds in 2013 and the figure is projected to rise to 200.235 tons by 2030. Although antimicrobials are used in comparable amounts in animals and humans (118 mg/PCU and 133 mg/kg respectively), the chances of mutations in animals are higher due to larger animal biomass (Van Boeckhel et al., 2017).

In Laos, where is weak or non-existent regulatory frameworks governing AMU, sub-optimal enforcement and compliance with existing guidelines, low levels of AMR awareness, and inadequate commitment to responsible antimicrobial stewardship are driving development of AMR. Other than, there is gaps in data and information across the food and agriculture sectors, therefore hamper the assessment of AMU and creating difficulties in determining levels of antimicrobial resistance (Poupaud et al., 2021).

3. Research Questions

1) What is the overall point prevalence of Eimeria species in poultry farms within the target area?
2) Which specific species (e.g., E.tenella, E.acervulina, E. maxima) are most dominant, and how does their distribution vary across different types of poultry production (e.g., broilers vs. layers)?
3) What environmental and management factors (such as litter moisture, stocking density, or seasonal changes) are most significantly associated with high oocyst counts?
4) Is there a correlation between the age of the fowl and the intensity of oocyst shedding in the sampled farms?

4. Hypothesis

1) Know kind of Coccidia infection in backyard native chicken flocks' production existed with the high prevalence in Vientiane of Laos?
2) The prevalence and species diversity of Eimeria will be significantly higher in intensive broiler operations compared to free-range systems due to higher stocking densities.

5.Objectives

1) To study the prevalence of Coccidia in the target area.
2) To study the anticoccidial resistance of Coccidia.

6. Sample Collection

Sampling Strategy: Collect fecal samples (litter or fresh droppings) from various farm types (broiler, layer, backyard) across different age groups.

7. Experimental Treatments

Grouping Treatments
Divide your fowl (usually day-old chicks raised in a coccidia-free environment until about day 14) into the following treatment groups:

Control Groups
Negative Control: Uninfected, Untreated Control. These birds receive no parasites and no drugs. This measures the "normal" growth potential of your birds.

Positive Control: Infected, Untreated Control. These birds are infected with your field isolates but receive no medication. This shows the full damage the parasite can do.

Testing Groups (The Treatments)
For each drug wish to test, create an Infected, Treated (IT) group. Common drugs to include are:

Chemical Treatment: Amprolium. (Known for rapid action but fast resistance development).

8. Test sample

1) Fowl is the chosen test animal
2) Usually, day-old chicks raised in a coccidia-free environment until about day 14 into the treatment groups

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